



**Politecnico
di Torino**

Politecnico di Torino

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CFD analysis of Alfa Romeo MiTo and evaluation of the effect of a front wheel spats on the aerodynamic performance

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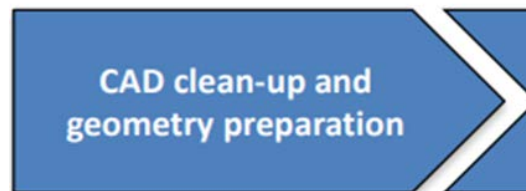
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REPORT 1- Aerodynamic performance of Alfa Romeo MiTo with CFD

The aim of the report is the Evaluation of aerodynamic performance of the Alfa Romeo MiTo with CFD simulation on StarCCM+ software and its comparison with experimental results. To do that, a standard workflow of activities has been performed:

- **Pre-processing phase** where the CAD model is received and the meshes are generated.
- **Setup simulation and run solver** is the part in which simulation is carried out.
- **Post-processing phase** where results are analyzed.

As conclusion, flow field interpretation and aerodynamics coefficients of drag and lift are reported, and then they are compared with experimental coefficients.

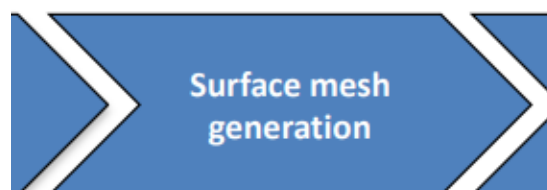


The input for CFD practice consists of a CAD model of Alfa Romeo MiTo which has been prepared to be compatible with the tools involved during the simulation.

More precisely, the following aspects have been already examined during a pre-processing phase:

- All the adjacent surfaces are connected.
- No penetration between different faces has been detected.

For this specific study case only half of the car geometry is simulated to halve computation times related to the solving of the air domain.



The simulation is approached through the discretization of the car surfaces.

Since a good compromise must be found between accuracy of the results and processing time, a finer mesh is defined for more sensitive aerodynamic details, such as the rear spoiler and diffuser area, while a coarser mesh is used for regions which are far from the vehicle.

Setup default controls have been provided for the mesh and then applied everywhere unless finer controls are specifically required.

Table 1: Default surface mesh settings

Parameter	Value	Description
Base size	25 mm	Reference length value for all relative size components. This serves as the baseline size for mesh cells.
Target surface size	200%	Defines the intended size of mesh cells on surfaces in areas without smaller constraints. A larger percentage indicates coarser cells, useful for regions of lower importance.
Minimum surface size	100%	Specifies the smallest allowable mesh cell size on surfaces. Cells below this size are eliminated to optimize computation and avoid unnecessary refinement.
Mesh geometry	Triangular	Specifies the shape of the mesh cells. In this case, the cells have a triangular geometry.
Mesh density	0.7	Defines the density of volumetric mesh cells in the domain, influencing the total number of elements.
Growth factor	1.3	Determines the rate of cell size transition between coarse and fine mesh regions, ensuring smooth transitions.

Once the default parameters are fixed, some customized dimensions are set with surface controls to improve the quality results without increasing computation time.

More precisely, 4 control surfaces have been customized as it follows:

Table 2: Customized surface mesh settings

Element	Target size	Minimum size
Vehicle	200%	100%
Side mirror, rear spoiler, taillight, wheels	75%	37.5%
Wind tunnel floor	2500%	250%
Wind tunnel other surfaces	5000%	250%

A valuable consideration concerns the amount of these percentages: the higher is the target size, the bigger are going to be the mesh dimensions. This means that, with respect to vehicle discretization, finer mesh is attributed to side mirror, rear spoiler, taillight and wheels while coarser one is related to tunnel floor and other surfaces.

Anisotropic mesh near the wall (“prism layer”) has been adopted to catch high gradients in the boundary layer region. The choice of the prism layer mesh is also affected by the solver that is going to be used: for this specific application, High Y+ approach would be used with already implemented-wall functions since high computational cost (for the evaluation of the boundary layer transitions) can’t be afforded.

While customizing the surface of the tunnel floor, a customization is activated also for “prism layer” with following values:

Table 3: Layer mesh settings

Parameter	Base prism layer mesh	Tunnel floor prism layer mesh
Prism layer near wall thickness	2 mm	4 mm
Number of layers	3	5
Total layer thickness	10 mm	50 mm

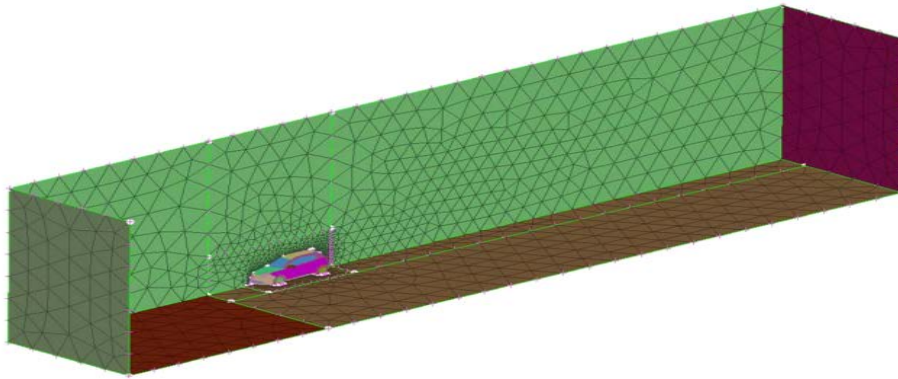
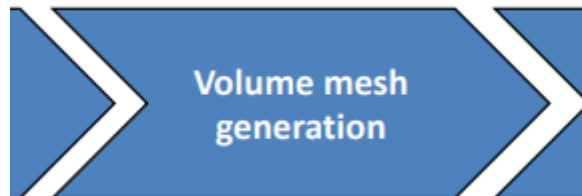


Figure 1: Wind tunnel surfaces discretization



The space around the body surfaces is also discretized: volumetric cells are created to evaluate, for each point of the space, the value of different properties such as velocity, temperature and pressure in post-processing phase.

Box shapes are then created: they are intended as control volumes that are going to be customized.

Table 4: Customized volume mesh settings

Volume Regions	Target Size	Description
VR-components	75%	Volume around spoiler, mirror, diffuser, and pillar
VR-vehicle wake	200%	Volume on the rear of the car where the wake is generated
VR-vehicle 1	500%	Volume in which the car is fitting
VR-vehicle 2	1000%	Volume in which the car is fitting + volume behind

For visualization purposes, a symmetrical plane has been created (parallel to the wind tunnel lateral surface): this allows to observe how the space around the model has been discretized.

Proceeding with the generation of the mesh with the features described until now, a total number of 315151 volume cells is assigned from the software to the model and the space around it.

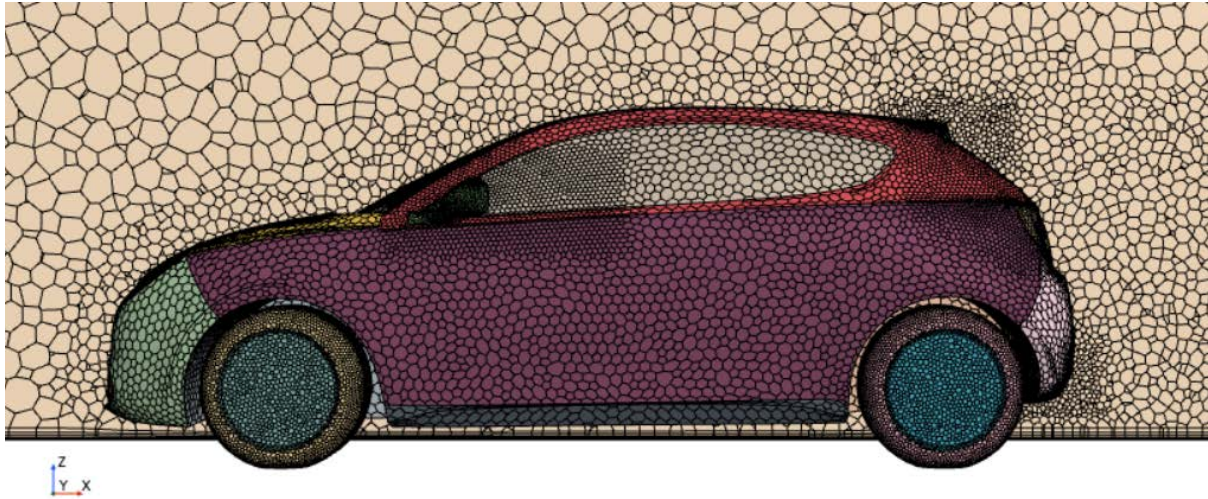
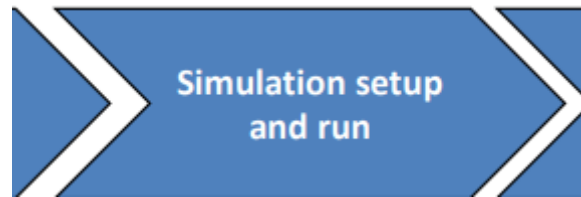


Figure 2: Mesh generation with a focus on customized details

The mesh generation is satisfactory since the number of volume cells is lower than 500000.



Physics Setup
1. Three dimensional simulation
2. Steady RANS analysis
3. Gas: air
- Density (ρ) = 1.225 kg/m ³
- Dynamic viscosity (μ) = 1.79×10^{-5} Pa/s
4. Segregated flow
5. Constant density (approximation for Mach number < 0.3)
6. Turbulent flow (with initial speed $u = 20$ m/s along x)
7. K-epsilon model (for closure problem)
8. Cell quality remediation (optional parameter to increase stability of the solution in case of bad cells)

Table 5: Simulation setup

For this application, a steady RANS analysis with k-epsilon turbulence model is adopted. This affects the simulation setup.

Boundary conditions need to be specified, concerning wind tunnel surfaces.

Boundary Conditions	Description
Wind Tunnel INLET	Velocity = 38.89 m/s
Wind Tunnel OUTLET	Pressure = 0 Pa
Wind Tunnel FLOOR	Front: Slip condition Rear: No slip condition
Wind Tunnel SIMMETRY	Symmetry plane
Wind Tunnel TOP	Slip condition
Wind Tunnel SIDE	Slip condition
All other boundary	No slip wall (default)

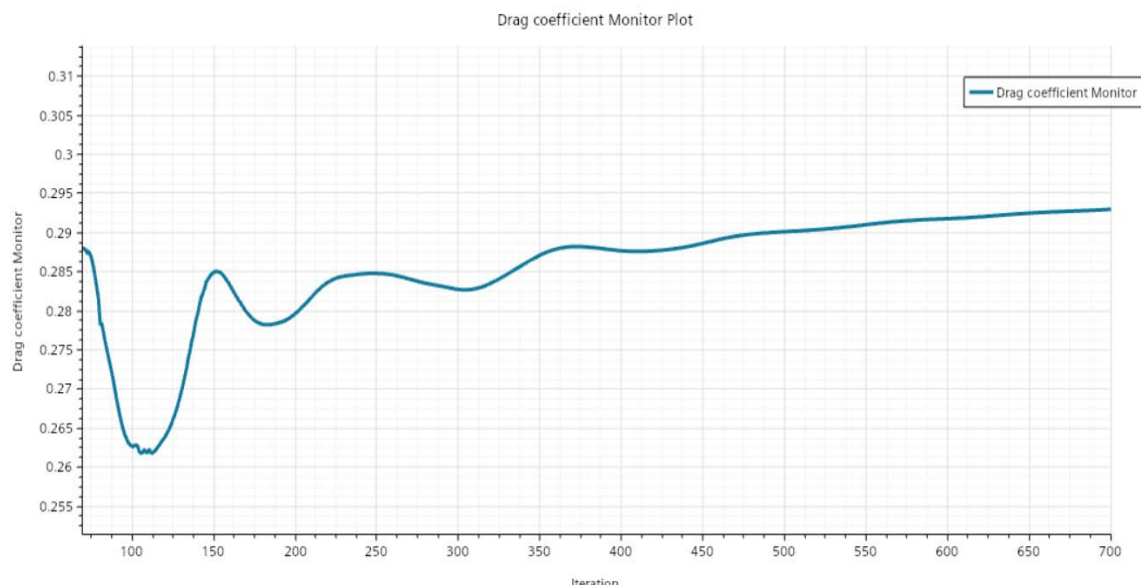
Table 6: Boundary conditions

The solver is set to stop after 700 iterations: it will provide lift and drag coefficients, and the residuals related to the computation.



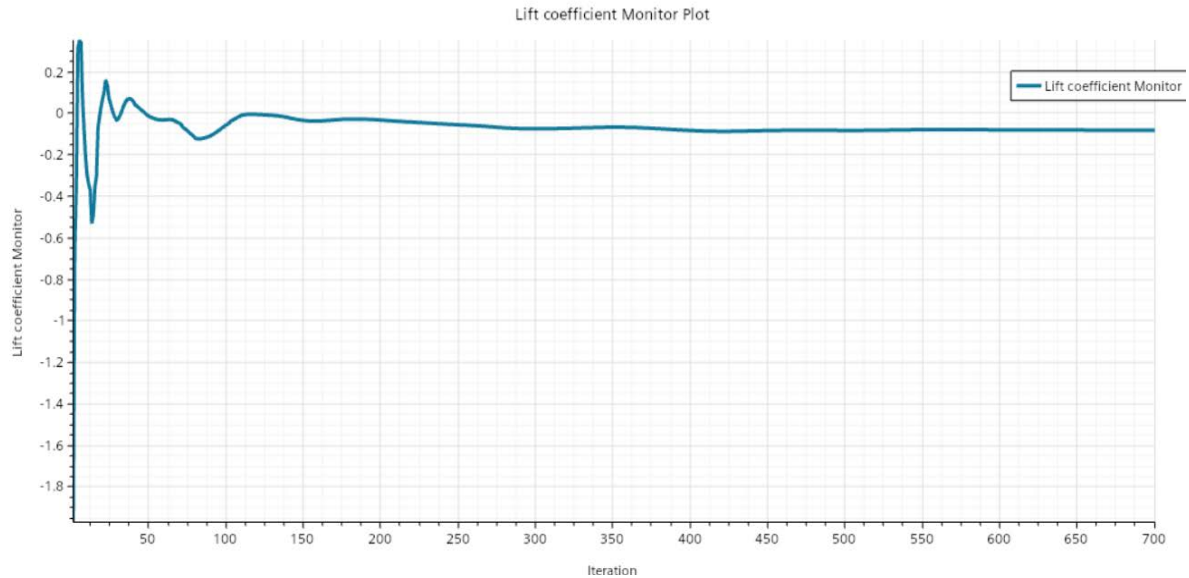
➤ Aerodynamic coefficients of drag and lift (C_D , C_L) and residuals

The drag coefficient plot shows a narrow oscillation range, suggesting a result which is not way too far from the final one, after a short number of iterations. The final value of the coefficient, estimated after 700 iterations, is $C_D = 0.293$.

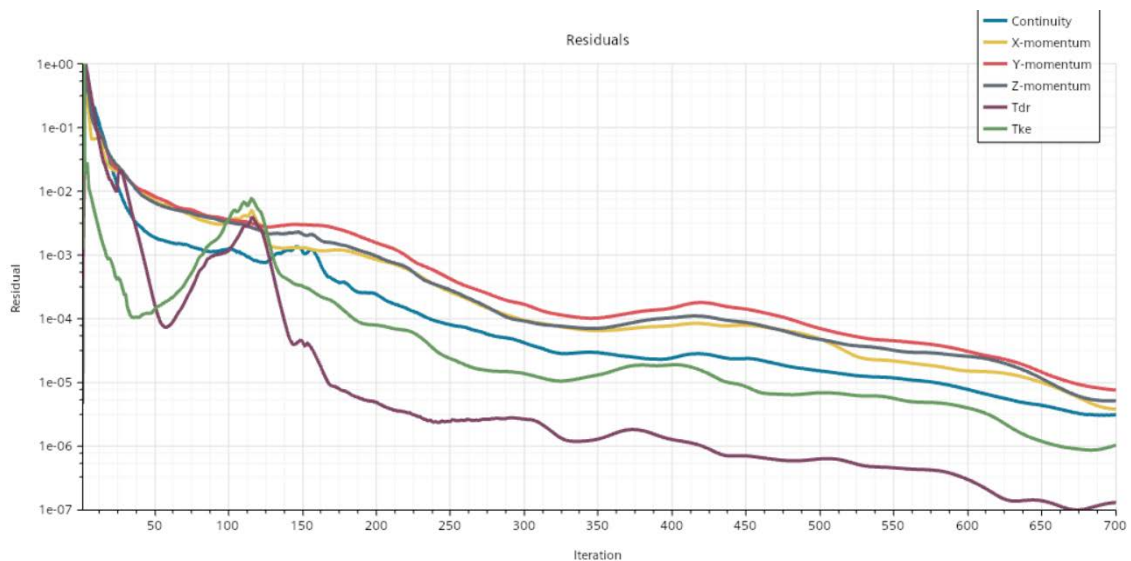


The lift coefficient plot shows a wider amplitude of the initial oscillation (with respect to the drag plot), suggesting a less precise result for a short number of iterations. The final value of the coefficient, estimated after 700 iterations, is $C_L = -0.103$.

The negative value of the lift coefficient is due to the downforce on the vehicle, related to underbody flatness, which is a simplification of the real model.



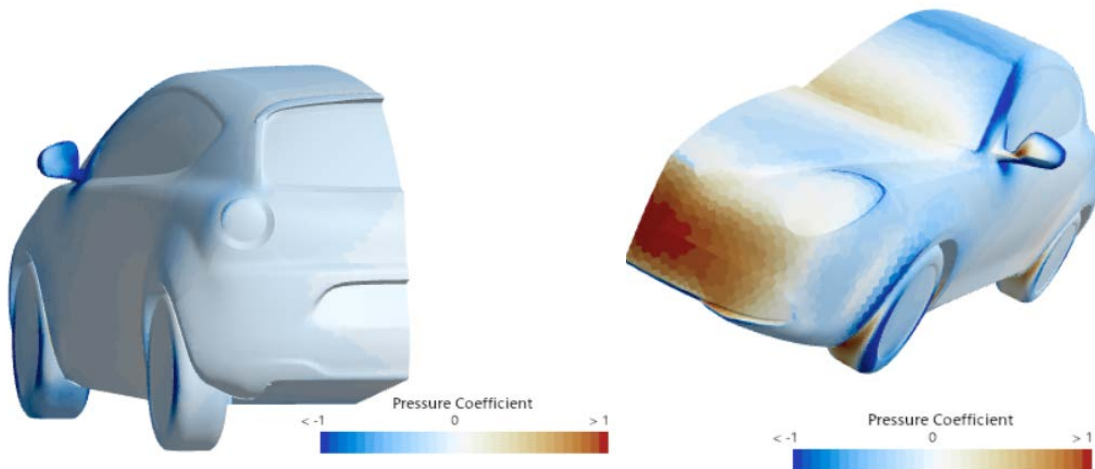
Residuals are an indicator of numerical solution convergence at each iteration. A converging simulation is obtained when residuals tend to zero: in second instance, this means that drag and lift coefficients values must oscillate in an extremely narrow range.



The residuals reported above confirm that the solution is acceptable: starting from the 500th iteration, their magnitude is lower than 10^{-4} (usually assumed as acceptable for industrial simulations).

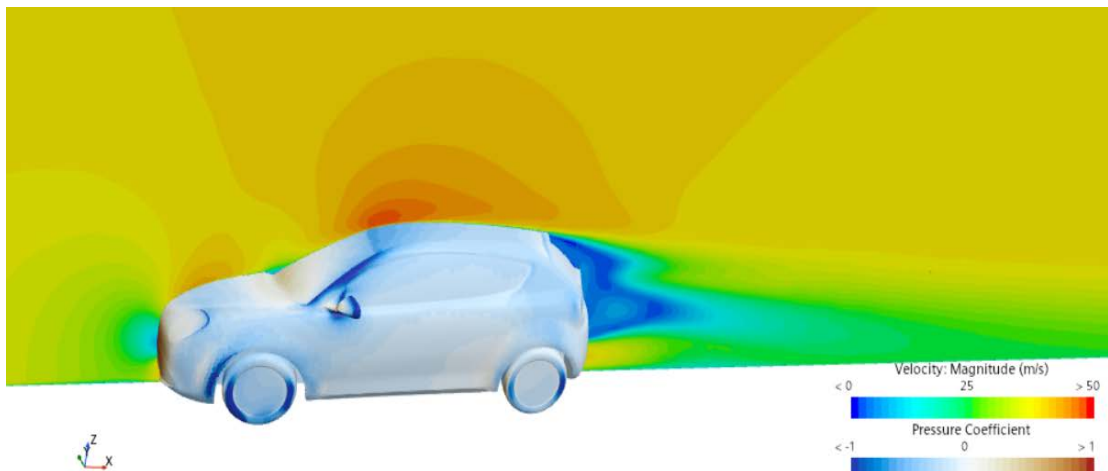
➤ **Flow field visualizations and interpretation**

Several different scalar field can be applied to bail out distinct characteristics of the flow field. One of these is the pressure coefficient C_p .



As expected, front part of the car causes the flow separation, while the rear portion of the car enlarges the wake.

It can be observed that the drag pressure on the front is generated in correspondence of the front grid (the most vertical part of the car with respect to the air flow) and on the lowest part of the windscreen.



In the same representation, also the magnitude of the velocity can be reported in order to capture some crucial details:

- Stagnation point is located where the velocity field reaches his zero because the fluid is drastically slowed down due to the presence of the frontal area of the vehicle.
- Separation point is located on the back of the vehicle, near the tip of the spoiler (where the wake is mainly generated).

Considering undisturbed conditions, far from the car, the colour map shows tone of light orange (corresponding to $u = 38.89$ m/s, which is the test condition fixed for the CFD and wind tunnel airflow of 140 km/h).

➤ **Comparison with EXP coefficients and suggestions for simulation methodology improvement**

A relative error coming from a comparison between wind tunnel measurements and CFD simulations can be now estimated:

$$\text{Wind tunnel results: } C_D = 0,290 \Rightarrow \varepsilon = \frac{0,293 - 0,290}{0,292} = 1\%$$

$$\text{Wind tunnel results: } C_L = 0,100 \Rightarrow \varepsilon = \frac{0,100 - 0,103}{0,100} = 3\%$$

Relative error committed on C_D is very low and well within acceptable margins for industrial applications. No further improvements are specifically required for C_D .

The relative error related to C_L is higher, indicating a substantial discrepancy between CFD results and experimental data. This value may affect the reliability of predictions regarding the vehicle's dynamic behaviour, especially in conditions where downforce plays a critical role.

It must be noted that wind tunnel simulations are also equipped with moving ground devices and rotating wheels which are able to return more accurate solution for lift simulations.

Proposed Improvements

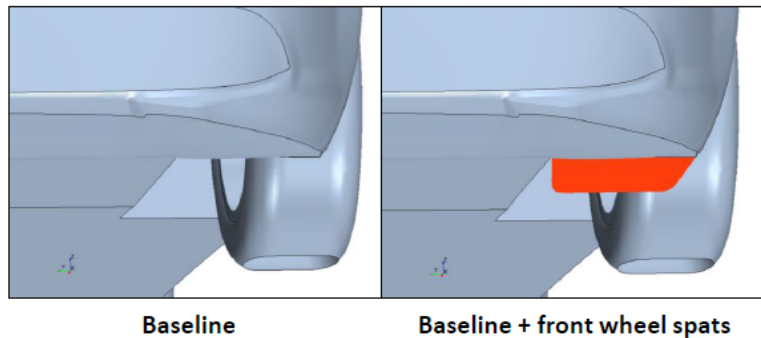
To reduce the error for C_L and improve the overall accuracy of the model, the following measures could be taken into account:

1. **Mesh refinement:** An increment on the mesh resolution in the lower part of the vehicle and in flow separation regions (e.g., spoiler and diffuser) could help to better capture critical aerodynamic details.
2. **Turbulence model:** more advanced turbulence models (such as DES - Detached Eddy Simulation) could improve the prediction of flow separation and vortex dynamics in the wake.
3. **Transient simulation:** for a better capture of dynamic flow fluctuation, which may not be adequately represented in steady-state simulations.

REPORT 2 - Evaluation of the effect of a geometric variant on the aerodynamic performance of the Alfa Romeo MiTo

Two different configurations of the Alfa Romeo MiTo are taken into account: the baseline, which has been previously discussed and analyzed for report 1, and a modified baseline with the addition of front wheel spats.

The aim of this report is to understand how this geometrical component affects the drag coefficient C_D .



The two models are characterized by the same type of volume mesh (trim mesh) for this comparison: the mesh is made of cubic elements.

For this simulation, a rolling road simulation system (RRS) is considered: some boundary conditions are applied on the velocity profile of the wheels in order to simulate their rotation. According to the velocity profile of the wheels, the ground shares the velocity of the lower point of the wheels (about 38.89 m/s considering the test conditions of the car).

Additionally, a moving reference frame for the rim spokes is considered.

Looking at the residuals of both the models, a convergence ratio of 10^{-5} can be observed at the end of the iteration process, also remembering that a satisfying amount for industrial cases is usually lower than 10^{-4} (as previously mentioned for report 1).

From a force convergence check on the drag coefficient on both models, it can be observed that results are stable and convergent (without any oscillatory behaviour). At the end of the iteration process:

Model	C_D
Baseline	0.278
Baseline + Front Wheel Spats	0.274

From the CFD simulation, it can be observed a difference of $\Delta C_D = -0.004$ meaning that a gain in terms of drag recovering can be achieved using the front wheel spats.

Since the comparison of the two models on wind tunnel simulations leads to a $\Delta C_D = -0.006$, it is noticeable that the CFD accuracy could be improved to get closer to the wind tunnel simulation (e.g. improving the mesh resolution, improving the quality of the surfaces which are not perfectly flat on the underbody, or even adopting a transient simulation).

Flow field visualization and interpretations

➤ Pressure coefficient C_p

$$C_p = \frac{p - p_\infty}{\frac{1}{2} \rho_\infty V_\infty^2}$$

Proprietà	Valore
Reference Density (ρ_∞)	1.225 kg/m ³
Reference Pressure (p_∞)	0.0 Pa
Reference Velocity (V_∞)	38.89 m/s

The surface pressure coefficient is useful to identify drag and lift generation regions.

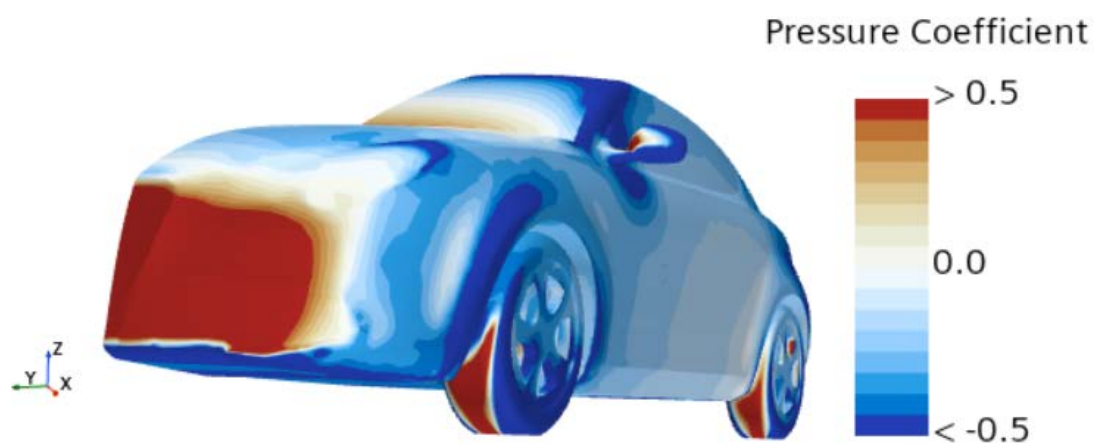


Figure 3: Baseline

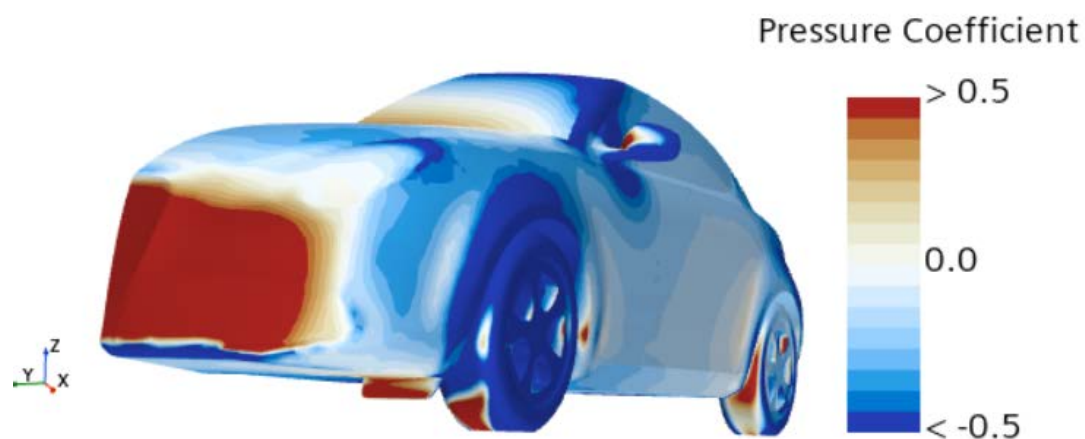


Figure 4: Baseline + front wheel spats

The flow field on the front of the car shows different types of cells:

- cells where $C_p > 0$ (e.g. frontal area of the bumper, wheels, side mirrors and lower part of the windscreen) $\Rightarrow$ air motion is causing a force in the opposite direction of the motion, generating drag

- cells where $C_p < 0$ (e.g. hood, pillars and higher part of the windscreen) $\Rightarrow$ the component of the forces acting on the x direction is oriented in the same direction of the motion, recovering drag.

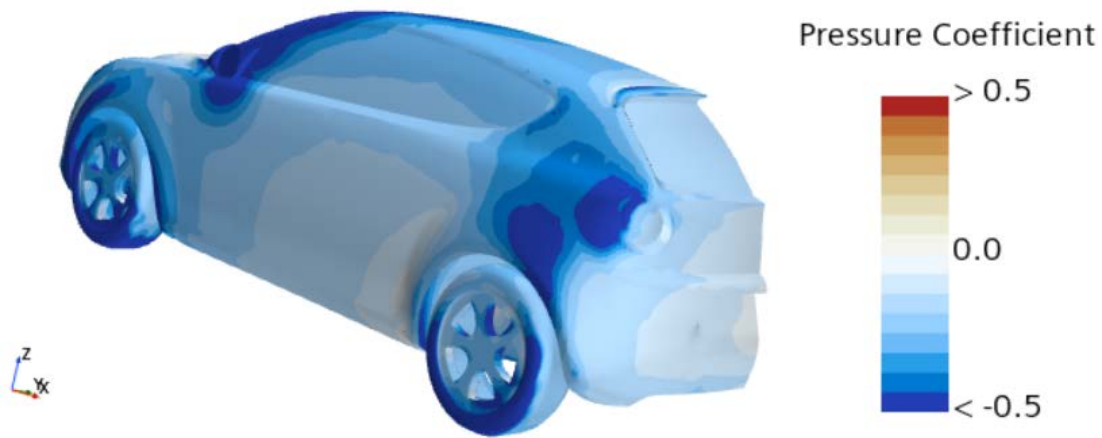


Figure 5: Baseline



Figure 6: Baseline + front wheel spats

In the back of the car, cells of surfaces are characterized by $C_p < 0 \Rightarrow$ generation of drag and wake.

In the **baseline** configuration, a larger area of positive pressure can be seen on the front wheel, indicating higher pressure accumulation.

With the **addition of front wheel spats**, overpressure is created directly on the spats. There is a noticeable reduction in the intensity of positive pressure on the tyres, suggesting that the wheel spats more effectively divert the airflow, also reducing pressure build-up at the front. Additionally, the wheel spats seem to increase the areas of low pressure near the front wheels (blue zones), improving the airflow distribution around these regions and reducing induced drag. A small intrusion of the flow inside the wheel arch can be observed in *figure 4*, causing a local worsening of the flow but globally, due to the spats, the flow changes also on the rear wheels profile with a pressure reduction (drag recovering).

➤ **Skin friction coefficient C_f**

$$C_f = \frac{\tau_w}{\frac{1}{2} \rho_\infty V_\infty^2}$$

The wall shear stress is expressed as $\tau_w = \mu \frac{\partial u}{\partial y}$ and normalized with respect to the reference values previously mentioned.

For standard passenger cars, the predominant part of the drag is related to the pressure, while a lower contribution (about 10%) is depending on friction.

The skin friction coefficient helps to visualize all the part of the vehicle that are characterized from the separation of the flow. Since this condition is expressed from $\frac{\partial u}{\partial y} = 0$ in correspondence of the surface (and $C_f \propto \tau_w = \mu \frac{\partial u}{\partial y}$), the separation of the flow occurs for all the cells where $C_f = 0$.

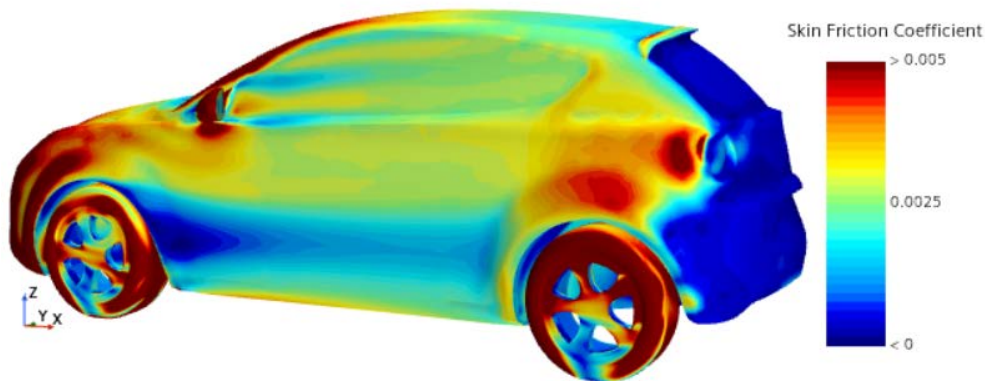


Figure 7: Baseline

For the baseline configuration, the wake region generated downstream the front wheel is contributing to the undesired effect of flow separation on the side of the car.

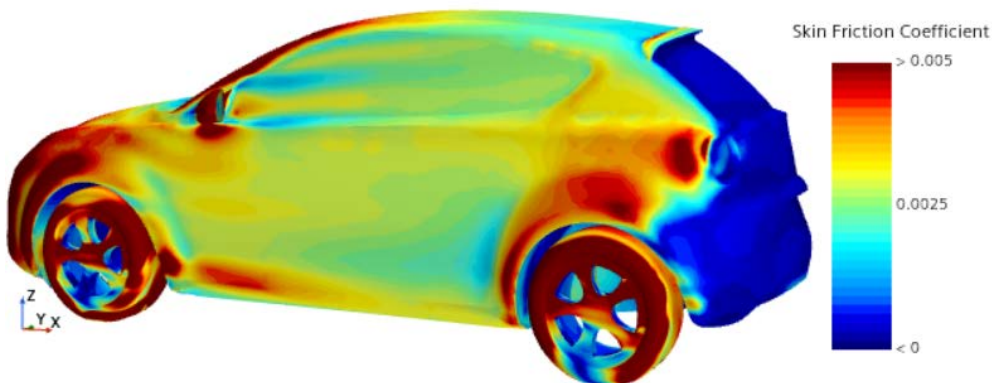


Figure 8: Baseline + front wheel spats

Introducing the spats, the flow separation on the side of the car is enormously reduced, improving the drag recovering on the central portion of the car.

➤ **Velocity flow field**

The velocity flow in a plane parallel with respect to the ground (with coordinate along z not far from the wheels) is considered.

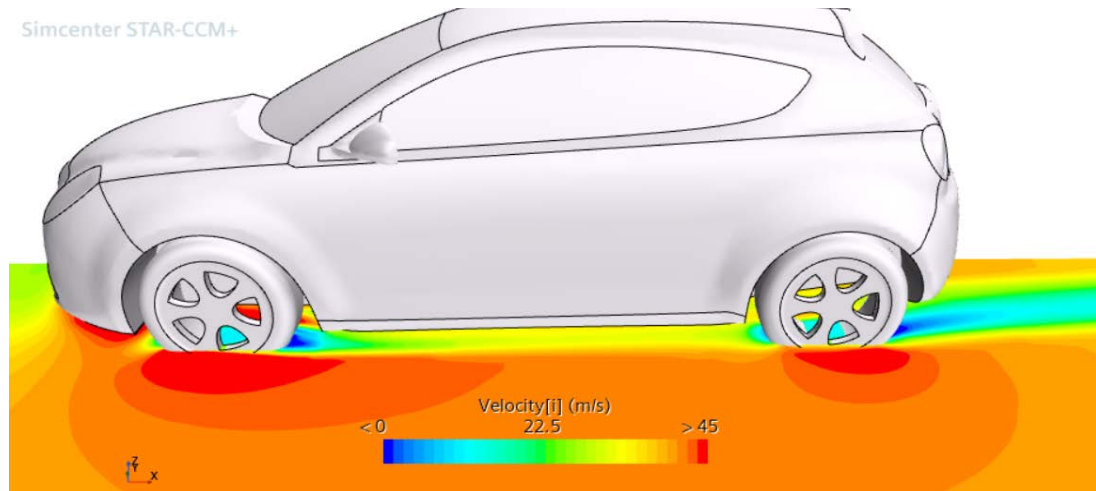


Figure 9: Baseline

The following features can be observed:

- strong acceleration of the flow in the region of the wheels;
- relevant wake generation behind the wheels (especially rear ones).

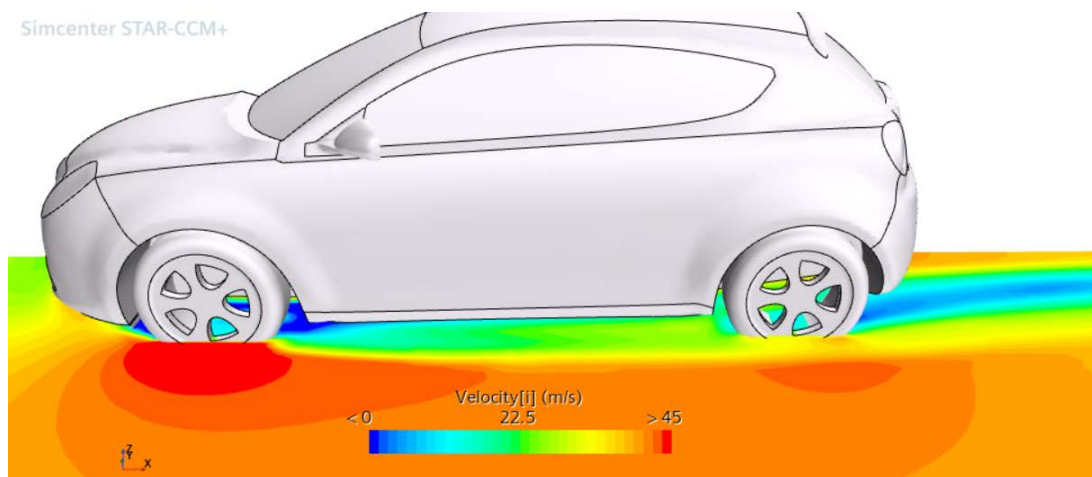


Figure 10: Baseline + front wheel spats

It can be observed that, adopting the spats, the acceleration of the flow in proximity of the rear wheels is slightly reduced (from the tonality of red).

➤ **Total pressure coefficient $C_{p,tot}$**

$$C_{p,tot} = \frac{p^{tot} - p_{\infty}}{\frac{1}{2} \rho_{\infty} V_{\infty}^2}$$

Total pressure coefficient is helpful to verify the area of the vehicle where negative total pressure is obtained due to energy losses, also detecting where flow separation and wake generation occur (assigning to them a proper extension).

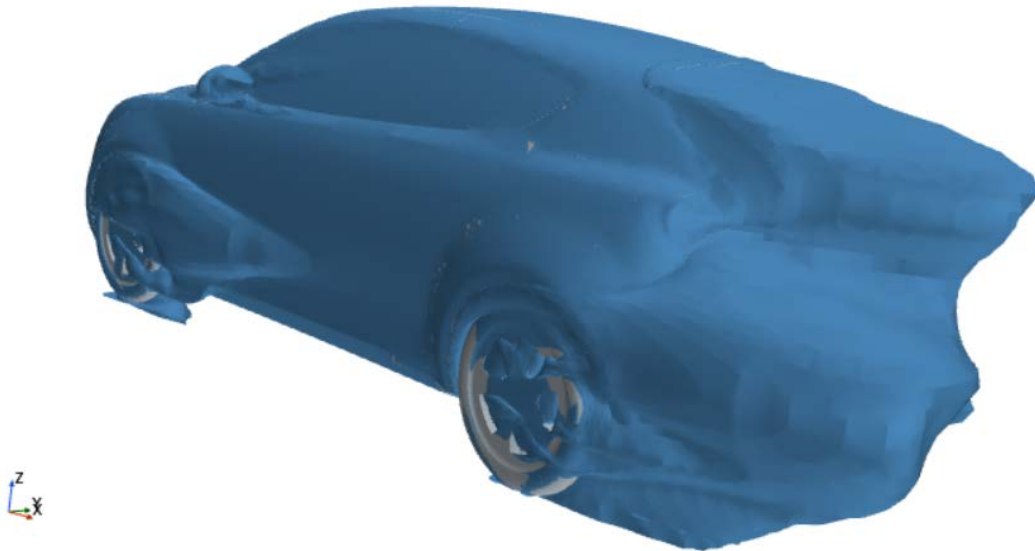


Figure 11: Baseline

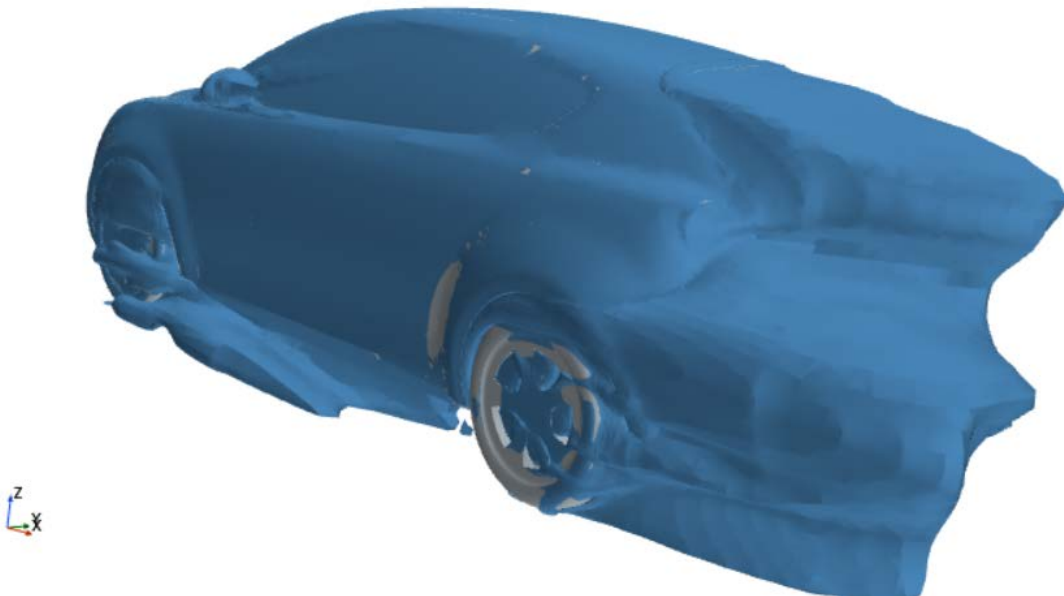
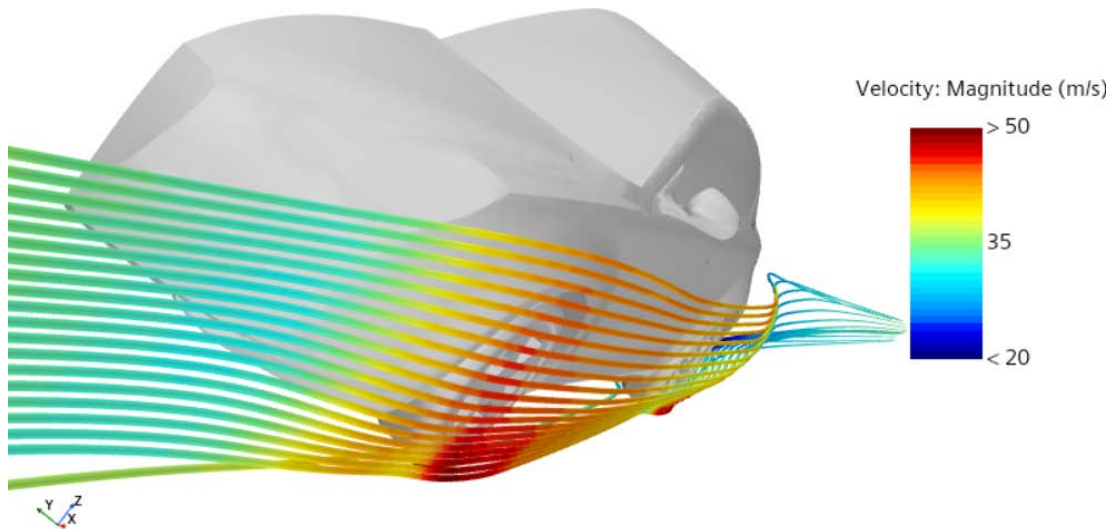


Figure 12: Baseline + front wheel spats

➤ Streamline

It is useful to identify flow field development



A better visualization of the behaviour of the flow impacting the front wheel (which is the area where the comparison between the two configurations are compared) can be obtained if the seed line is differently placed: this choice would allow to capture inner streamlines, enlightening the different impact of the configuration with spats with respect to the baseline.

Placing the seed line on a coordinate $y = -0.5\text{m}$ in front of the wheel and looking at the car from the bottom:

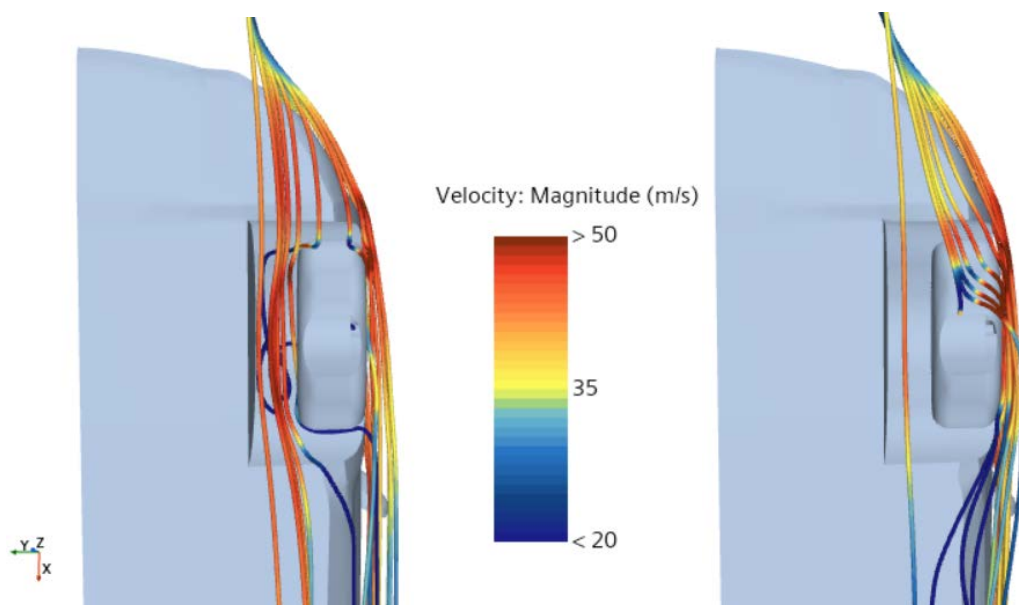
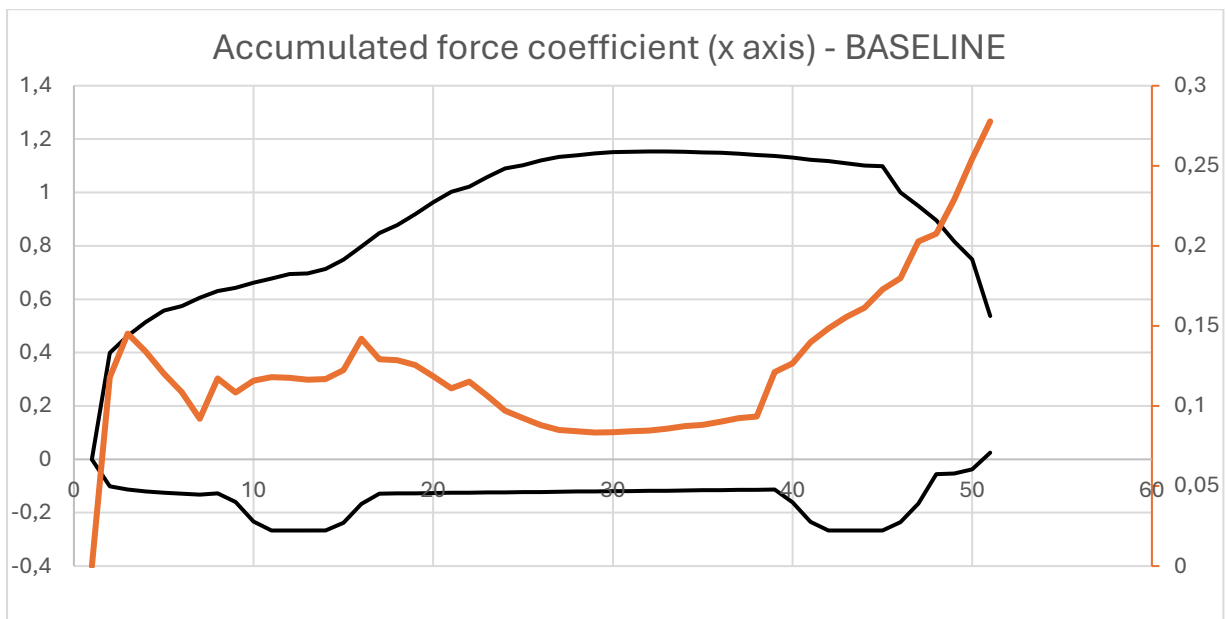


Figure 13: Baseline (left) and baseline + front wheel spats (right) streamline

Looking at the baseline model on the left, the flow directly impacts with the tyre while, on the right, the spat helps deflecting the flow on the outside of the wheel arch.

➤ Drag development on x axis

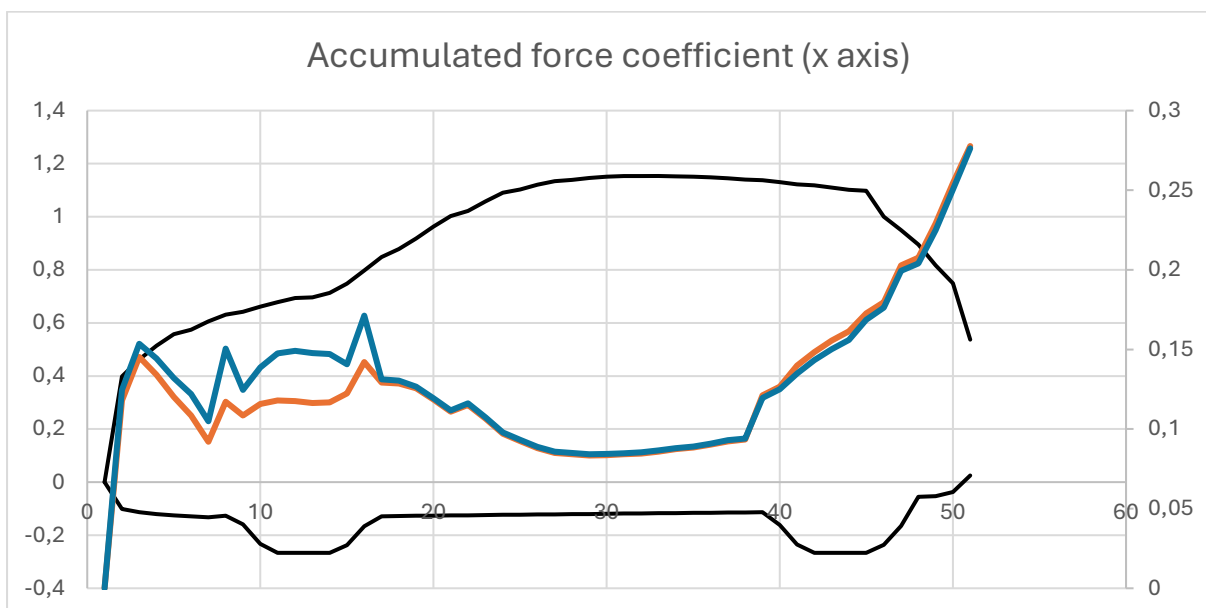


Analysing the chart from the front to the back of the car profile, a huge amount of drag is generated from the frontal part of the bumper region. Negative pressure on the hood allows a recovery of drag but, after the first minimum, a spike can be observed due to the drag generated from the front wheels profile. Drag still increases in correspondence of the lower part of the windscreen and where side mirrors are located since these areas are characterized by a overpressure already mentioned when discussing the pressure coefficient.

Reduction of the drag is achieved with a consistent negative pressure on the roof and all the flat surfaces (also on the bottom of the car), increasing once again on the back of the car where the wake is generated.

The cumulative value of the drag on the car profile leads to a result of $C_D = 0.278$.

Considering both the configurations of the Alfa Romeo MiTo:



Since the spats are mounted on the front wheels, the main differences in terms of drag can be observed on the front of the vehicle. The presence of the spats (blue line) is locally generating more drag on the front with respect to the model without them (orange line). However, their beneficial effects can be detected behind the front wheels, where a reduction in terms of wake and flow separation on the lateral side has been previously discussed.

Therefore, the drag locally generated on the front is globally recovered on the back of the car with a cumulative value of $C_D = 0.274$.